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import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, StandardScaler

# 1. Load the Titanic dataset and convert it into a DataFrame
# We'll load the dataset from seaborn's built-in datasets
df = sns.load_dataset('titanic')
display(df.head())
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male	deck	embark_town	alive	alone
0	0	3	male	22.0	1	0	7.2500	S	Third	man	True	NaN	Southampton	no	False
1	1	1	female	38.0	1	0	71.2833	C	First	woman	False	C	Cherbourg	yes	False
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	False	NaN	Southampton	yes	True
3	1	1	female	35.0	1	0	53.1000	S	First	woman	False	C	Southampton	yes	False
4	0	3	male	35.0	0	0	8.0500	S	Third	man	True	NaN	Southampton	no	True

```
# 2. Explore and Understand the Dataset
print("--- Dataset Info ---")
df.info()

print("\n--- Missing Values ---")
print(df.isnull().sum())
```

```
--- Dataset Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    object
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    object
 8   class        891 non-null    category
 9   who          891 non-null    object
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    object
 13  alive        891 non-null    object
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), object(5)
memory usage: 80.7+ KB

--- Missing Values ---
survived         0
pclass           0
sex              0
age            177
sibsp            0
parch            0
fare             0
embarked         2
class            0
who              0
adult_male       0
deck           688
embark_town      2
alive            0
alone            0
dtype: int64
```

```
# 3. Do forward, backward fill on Age
df['age'] = df['age'].ffill().bfill()
```

```
# 4. Fill missing values in Cabin with “unknown” and limit to 5
# In seaborn dataset, the column is 'deck'. If we use standard titanic sources, it is 'cabin'.
# Let's ensure both are handled or mapped correctly. We will define a 'cabin' column from deck or handle it if present.
if 'cabin' not in df.columns and 'deck' in df.columns:
    df['cabin'] = df['deck'].astype(str)

df['cabin'] = df['cabin'].fillna('unknown')
# Limit the display of unique values or cap them if needed; here we fill 'unknown'
df.loc[df['cabin'] == 'nan', 'cabin'] = 'unknown'
display(df['cabin'].head(5))
```

	cabin
0	unknown
1	C
2	unknown
3	C
4	unknown

dtype: object

```
# 5. Remove Duplicate Records if any
initial_shape = df.shape
df = df.drop_duplicates()
print(f"Removed {initial_shape[0] - df.shape[0]} duplicate records.")
```

```
Removed 56 duplicate records.
```

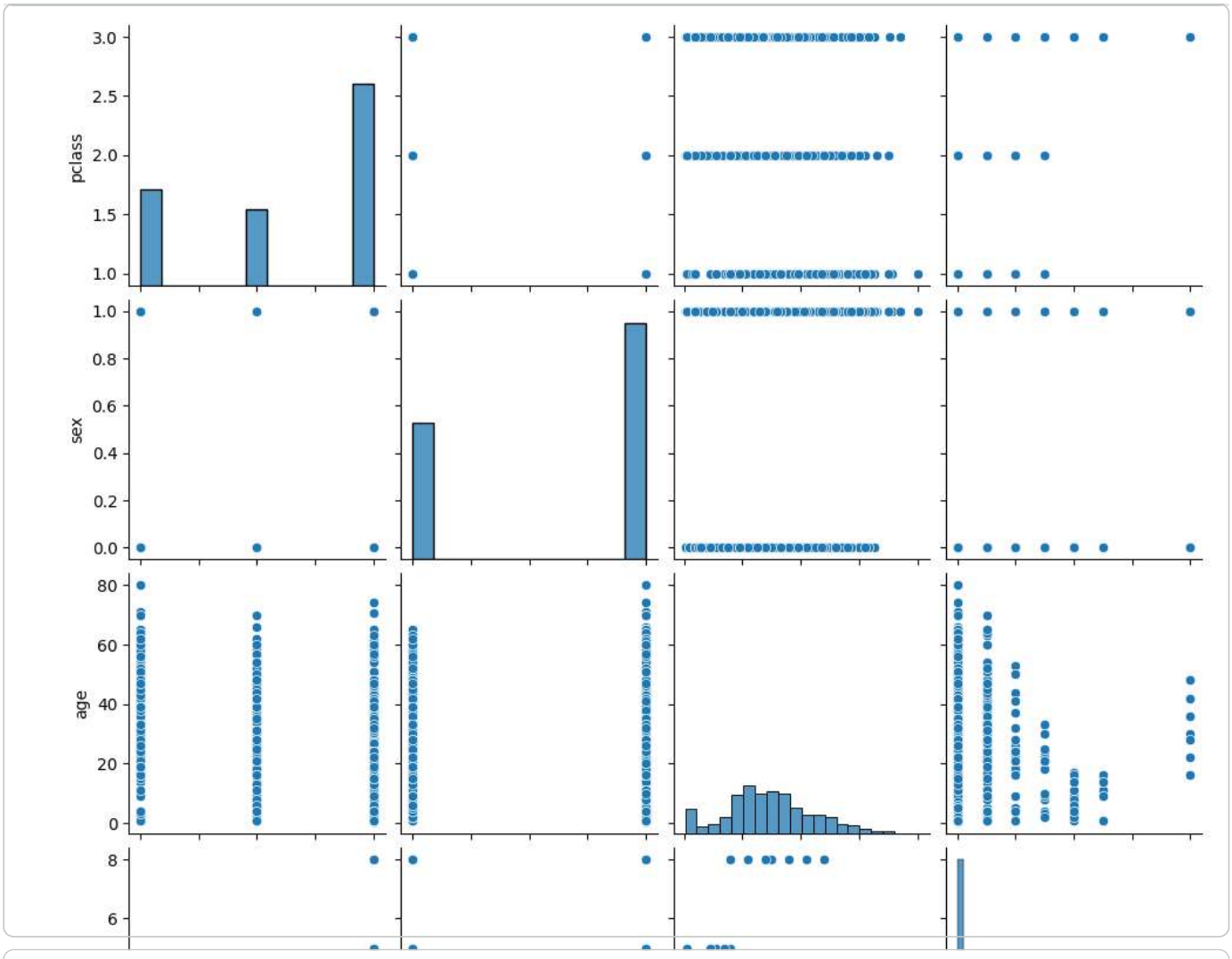
```
# 6. Encode Categorical Columns Sex using LabelEncoder
le = LabelEncoder()
df['sex'] = le.fit_transform(df['sex'])
display(df[['sex']].head())
```

	sex
0	1
1	0
2	0
3	0
4	1

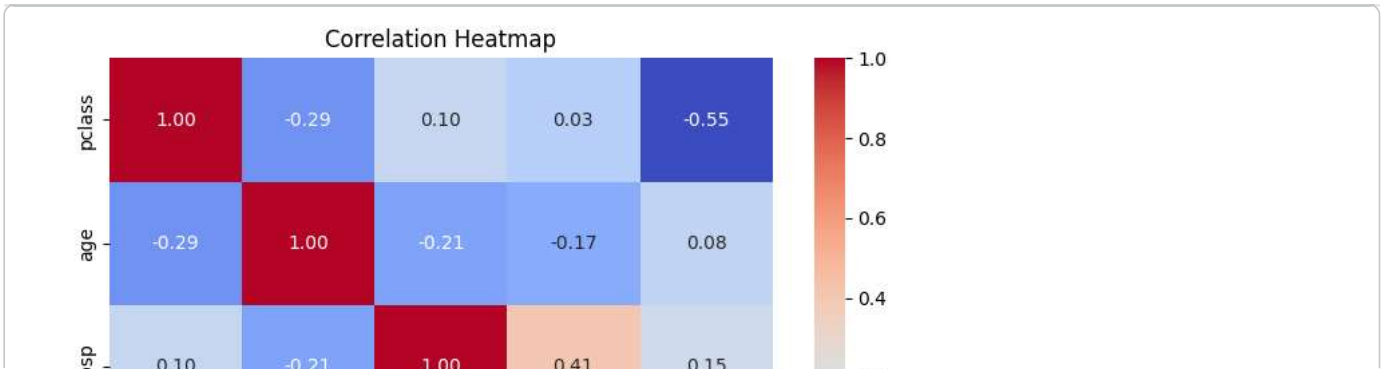
```
# 7. Scale Numerical Feature Fare using StandardScaler
scaler = StandardScaler()
df['fare'] = scaler.fit_transform(df[['fare']])
display(df[['fare']].head())
```

	fare
0	-0.516916
1	0.740208
2	-0.503664
3	0.383227
4	-0.501210

```
# 8. Pair Plot of Selected Features Pclass, Sex, Age, SibSp
# Adjust keys for seaborn titanic dataset columns: 'pclass', 'sex', 'age', 'sibsp'
sns.pairplot(df[['pclass', 'sex', 'age', 'sibsp']])
plt.show()
```




```
# 9. Display the Correlation Heatmap for Pclass, Age, SibSp, Parch, Fare
# Mapping lowercase column names standard in seaborn titanic
corr_features = ['pclass', 'age', 'sibsp', 'parch', 'fare']
plt.figure(figsize=(8, 6))
sns.heatmap(df[corr_features].corr(), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Heatmap')
plt.show()
```


Correlation Heatmap
pclass
age
sp
1.00
-0.29
0.10
0.03
-0.55
-0.29
1.00
-0.21
-0.17
0.08
0.10
-0.21
1.00
0.41
0.15
1.0
0.8
0.6
0.4